Toom-Cook

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Motivation + Overview

Divide and conquer algorithm used to multiply two very large integers

Turns $O(n^2)$ into $O(n^e)$ where $e = \log(2k - 1) / \log(k)$

Toom-3 is a single instance of the algorithm where $\boldsymbol{k = 3}$

Go-to choice for integers that are too large for Karatsuba but not yet large enough to justify complexity of Fast Fourier Transforms

Usage:

- Cryptography (RSA)
- Large integer libraries

Polynomials: example using 123,456 × 654,321

$$123{,}456 \times 654{,}321$$

$$12 \mid 34 \mid 56 \times 65 \mid 43 \mid 21$$

$$x_2 = 12,\ x_1 = 34,\ x_0 = 56 \times y_2 = 65,\ y_1 = 43,\ y_0 = 21$$

Now create the polynomials using base 100…

$$12(100)^2 + 34(100)^1 + 56(100)^0 \times 65(100)^2 + 43(100)^1 + 21(100)^0$$

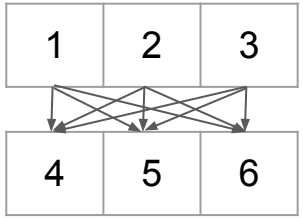
$$12\cancel{(100)}^2 + 34\cancel{(100)}^1 + 56\cancel{(100)}^0 \times 65\cancel{(100)}^2 + 43\cancel{(100)}^1 + 21\cancel{(100)}^0$$

We get…

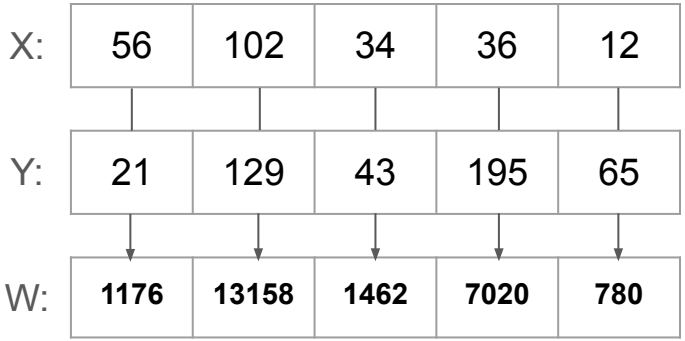
$$X(t) = 12t^2 + 34t + 56 \times Y(t) = 65t^2 + 43t + 21$$

Why is this better? Pointwise Multiplication and Recursion

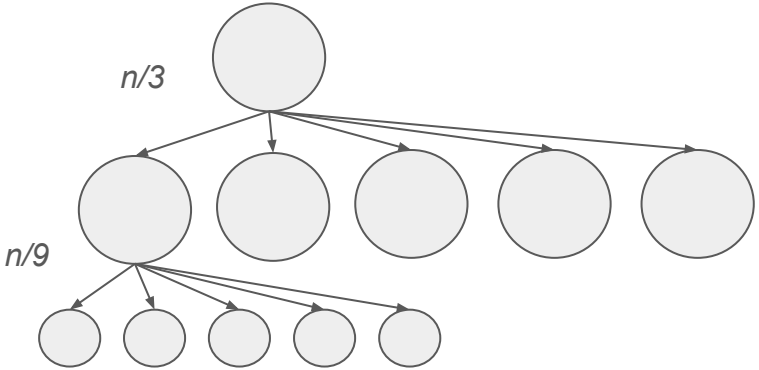
Classic multiplication with 9 multiplications:



Toom-Cook with 5 multiplications:



What if X(1) and Y(1) were *n* digits long?



Basically this and chunking by 3 (hence Toom-3):

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if len(X) < threshold:
    return X * Y
else:
    return toom_cook_multiply(X, Y)
```

Evaluation with Magic Points (0, 1, -1, -2, ∞)

Point (t)	$X(t) = 12t^2 + 34t + 56$	$Y(t) = 65t^2 + 43t + 21$	$W(t) = X \times Y$
t = 0	56	21	**1176**
t = 1	12 + 34 + 56 = 102	65 + 43 + 21 = 129	**13158**
t = -1	12 - 34 + 56 = 34	65 - 43 + 21 = 43	**1462**
t = -2	12(4) - 34(2) + 56 = 36	65(4) - 43(2) + 21 = 195	**7020**
t = ∞	12	65	**780**

This step (x*y for each t) would be evaluated recursively in the algorithm implementation, until small enough to be done naïvely

$$X(t) \times Y(t) = W(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 + c_4 t^4$$

$$W(0) = c_0 + \cancel{c_1(0)} + \cancel{c_2(0)^2} + \cancel{c_3(0)^3} + \cancel{c_4(0)^4} = c_0$$

$$W(1) = c_0 + c_1 + c_2 + c_3 + c_4$$

$$W(-1) = c_0 - c_1 + c_2 - c_3 + c_4$$

$$W(-2) = c_0 - 2c_1 + 4c_2 - 8c_3 + 16c_4$$

$$W(\infty) = \cancel{c_0} + \cancel{c_1()} + \cancel{c_2()^2} + \cancel{c_3()^3} + c_4 = c_4$$

Interpolation

Recall that:

$W(0) = c_0 = 1176$

$W(1) = c_0 + c_1 + c_2 + c_3 + c_4 = 13158$

$W(-1) = c_0 - c_1 + c_2 - c_3 + c_4 = 1462$

$W(-2) = c_0 - 2c_1 + 4c_2 - 8c_3 + 16c_4 = 7020$

$W(\infty) = \cancel{e_0} + \cancel{e_1()} + \cancel{e_2()^2} + \cancel{e_3()^3} + c_4 = c_4 = 780$

$$\begin{pmatrix} W(0) \\ W(1) \\ W(-1) \\ W(-2) \\ W(\infty) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & -2 & 4 & -8 & 16 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix}$$

But we still need c_1, c_2, and c_3 !

Invert like so:

$$\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & -1 & \frac{1}{6} & -2 \\ -1 & \frac{1}{2} & \frac{1}{2} & 0 & -1 \\ -\frac{1}{2} & \frac{1}{6} & \frac{1}{2} & -\frac{1}{6} & 2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} W(0) \\ W(1) \\ W(-1) \\ W(-2) \\ W(\infty) \end{pmatrix}$$

Extract our equations for c's$_{1\text{-}3}$:

$$c_1 = \frac{1}{2}W(0) + \frac{1}{3}W(1) - 1W(-1) + \frac{1}{6}W(-2) - 2W(\infty)$$

$$c_2 = -1W(0) + \frac{1}{2}W(1) + \frac{1}{2}W(-1) + 0W(-2) - 1W(\infty)$$

$$c_3 = -\frac{1}{2}W(0) + \frac{1}{6}W(1) + \frac{1}{2}W(-1) - \frac{1}{6}W(-2) + 2W(\infty)$$

Recomposition

Solving for $\mathbf{c_1}$, $\mathbf{c_2}$, and $\mathbf{c_3}$ by plugging in the W(t) from earlier like this:

$$c_1 = \frac{1}{2}W(0) + \frac{1}{3}W(1) - 1W(-1) + \frac{1}{6}W(-2) - 2W(\infty)$$

$$c_1 = \frac{1176}{2} + \frac{13158}{3} - 1462 + \frac{7020}{6} - 2(780)$$

we get:

$\mathbf{c_1}$= **3122**, $\mathbf{c_2}$= **5354**, and $\mathbf{c_3}$= **2726**

Recall that these are base-10 coefficients that are part of our polynomial:

$W(t) = c_0 + \mathbf{c_1}t + \mathbf{c_2}t^2 + \mathbf{c_3}t^3 + c_4t^4$

Remembering that $c_0 = 1176$ & $c_4 = 780$, we have:

$W(t) = 780t^4 + \mathbf{2726}t^3 + \mathbf{5354}t^2 + \mathbf{3122}t + 1176$

Now we need to put our base (**B** = **100**) back into W(t):

$W(\mathbf{B}) = c_0 + c_1(\mathbf{B}) + c_2(\mathbf{B})^2 + c_3(\mathbf{B})^3 + c_4(\mathbf{B})^4$

$W(B) = 1176 + 3122(\mathbf{100}) + 5354(\mathbf{100})^2 + 2726(\mathbf{100})^3 + 780(\mathbf{100})^4$

Then just add it all up:

```
            1,176
          312,200
       53,540,000
    2,726,000,000
+ 78,000,000,000
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   80,779,853,376
```

Try it on your calculators!

The End

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